

$$\left[\begin{array}{cccccccc}
0 & 0 & 0 & 0 & -\frac{\sqrt{2}i(2d_x^{12}+2id_y^{12}-d_x^{13}-id_y^{13}-d_x^{23}-id_y^{23})}{8} & 0 & \frac{\sqrt{6}i(d_x^{13}+id_y^{13}-d_x^{23}-id_y^{23})}{8} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i(2d_z^{12}-d_z^{13}-d_z^{23})}{12} & -\frac{\sqrt{6}i(2d_x^{12}+2id_y^{12}-d_x^{13}-id_y^{13}-d_x^{23}-id_y^{23})}{24} & \frac{\sqrt{2}i(d_z^{13}-d_z^{23})}{8} & \frac{\sqrt{2}i(d_x^{13}+id_y^{13}-d_x^{23}-id_y^{23})}{8} \\
0 & 0 & 0 & 0 & \frac{\sqrt{6}i(2d_x^{12}-2id_y^{12}-d_x^{13}+id_y^{13}-d_x^{23}+id_y^{23})}{24} & -\frac{\sqrt{6}i(2d_z^{12}-d_z^{13}-d_z^{23})}{24} & -\frac{\sqrt{2}i(d_x^{13}-id_y^{13}-d_x^{23}+id_y^{23})}{8} & \frac{\sqrt{2}i(d_z^{13}-d_z^{23})}{8} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i(2d_x^{12}-2id_y^{12}-d_x^{13}+id_y^{13}-d_x^{23}+id_y^{23})}{8} & 0 & -\frac{\sqrt{6}i(d_x^{13}-id_y^{13}-d_x^{23}+id_y^{23})}{8} \\
\frac{\sqrt{2}i(2d_x^{12}-2id_y^{12}-d_x^{13}+id_y^{13}-d_x^{23}+id_y^{23})}{8} & \frac{\sqrt{6}i(2d_z^{12}-d_z^{13}-d_z^{23})}{12} & -\frac{\sqrt{6}i(2d_x^{12}+2id_y^{12}-d_x^{13}-id_y^{13}-d_x^{23}-id_y^{23})}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i(d_x^{12}+id_y^{12}+d_x^{13}+id_y^{13}+d_x^{23}+id_y^{23})}{8} \\
0 & \frac{\sqrt{6}i(2d_x^{12}-2id_y^{12}-d_x^{13}+id_y^{13}-d_x^{23}+id_y^{23})}{12} & \frac{\sqrt{6}i(2d_z^{12}-d_z^{13}-d_z^{23})}{24} & -\frac{\sqrt{2}i(2d_x^{12}+2id_y^{12}-d_x^{13}-id_y^{13}-d_x^{23}-id_y^{23})}{8} & 0 & 0 & -\frac{\sqrt{3}i(d_x^{12}-id_y^{12}+d_x^{13}-id_y^{13}+d_x^{23}-id_y^{23})}{6} & -\frac{\sqrt{3}i(d_z^{12}+id_y^{12}+d_z^{13}+id_y^{13}+d_z^{23}+id_y^{23})}{6} \\
-\frac{\sqrt{6}i(d_x^{13}-id_y^{13}-d_x^{23}+id_y^{23})}{8} & -\frac{\sqrt{2}i(d_z^{13}-d_z^{23})}{4} & \frac{\sqrt{2}i(d_x^{13}+id_y^{13}-d_x^{23}-id_y^{23})}{8} & 0 & -\frac{\sqrt{3}i(d_z^{12}+d_z^{13}+d_z^{23})}{6} & \frac{\sqrt{3}i(d_x^{12}+id_y^{12}+d_x^{13}+id_y^{13}+d_x^{23}+id_y^{23})}{6} & 0 & -\frac{\sqrt{3}i(d_z^{12}+d_z^{13}+d_z^{23})}{6} \\
0 & -\frac{\sqrt{2}i(d_x^{13}-id_y^{13}-d_x^{23}+id_y^{23})}{8} & -\frac{\sqrt{2}i(d_z^{13}-d_z^{23})}{4} & \frac{\sqrt{6}i(d_x^{13}+id_y^{13}-d_x^{23}-id_y^{23})}{8} & \frac{\sqrt{3}i(d_x^{12}-id_y^{12}+d_x^{13}-id_y^{13}+d_x^{23}-id_y^{23})}{6} & \frac{\sqrt{3}i(d_z^{12}+d_z^{13}+d_z^{23})}{6} & 0 & 0
\end{array} \right]$$